Q1 (20 July 2021 Shift 1)

A certain charge Q is divided into two parts q and (Q-q). How should the charges Q and q be divided so that q and (Q-q) placed at a certain distance apart experience maximum electrostatic repulsion ?

(1) $\mathrm{Q}=\frac{\mathrm{q}}{2}$

(2) $\mathrm{Q}=2\mathrm{q}$

(3) $\mathrm{Q}=4\mathrm{q}$

(4) $\mathrm{Q}=3\mathrm{q}$

Q2 (22 July 2021 Shift 1)

An electric dipole is placed on x -axis in proximity to a line charge of linear charge density $3.0 \times 10^{-6} \mathrm{C}/\mathrm{m}$. Line charge is placed on z -axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4 N is exerted on the dipole, find out the amount of positive or negative

charge of the dipole.

(1) $815.1\mathrm{nC}$

(2) $8.8\mu\mathrm{C}$

(3) $0.485\mathrm{mC}$

(4) $4.44\mu\mathrm{C}$

Q3 (22 July 2021 Shift 1)

The total charge enclosed in an incremental

volume of $2 \times 10^{-9}\ \mathrm{m}^3$ located at the origin is ____ nC, if electric flux density of its field is found as $\mathrm{D}=\mathrm{e}^{-\mathrm{x}} \operatorname{siny} \hat{\mathrm{i}}-\mathrm{e}^{-\mathrm{x}} \operatorname{cosy} \hat{\mathrm{j}}+2\mathrm{z}\hat{\mathrm{k}} \mathrm{C}/\mathrm{m}^2$.

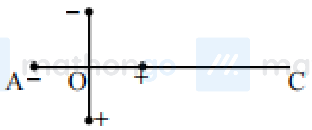
Q4 (25 July 2021 Shift 1)

A particle of mass 1mg and charge q is ly ing at the mid-point of two stationary particles kept at a distance ' 2 m ' when each is carrying same charge $'q'$. If the free charged particle is displaced from its equilibrium

position through distance ' x' (x << 1 m). The particle executes SHM. Its angular frequency of oscillation will be $\times 10^5 \mathrm{rad/s}$
if $q^2 = 10\mathrm{C}^2$

Q5 (25 July 2021 Shift 2)

Two ideal electric dipoles A and B, having their dipole moment p_1 and p_2 respectively are placed on a plane with their centres at O as shown in the figure. At point C on the axis of dipole A, the resultant electric field is making an angle of 37° with the axis. The ratio of the dipole moment of A and B, $\frac{\mathrm{p}_1}{\mathrm{p}_2}$ is : (take $\sin 37^{\circ} = \frac{3}{5}$)



(1) $\frac{3}{8}$
(2) $\frac{3}{2}$
(3) $\frac{2}{3}$
(4) $\frac{4}{3}$

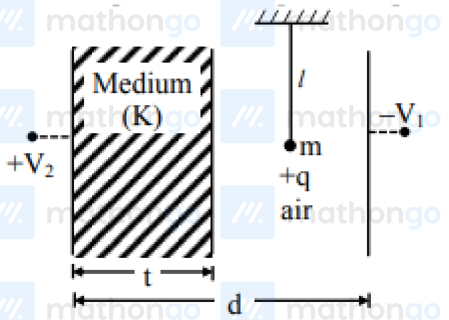
Q6 (27 July 2021 Shift 1)

The relative permittivity of distilled water is 81 . The velocity of light in it will be:
(Given $\mu_r = 1$)

(1) $4.33 \times 10^7 \mathrm{~m/s}$
(2) $2.33 \times 10^7 \mathrm{~m/s}$
(3) $3.33 \times 10^7 \mathrm{~m/s}$
(4) $5.33 \times 10^7 \mathrm{~m/s}$

Q7 (27 July 2021 Shift 2)

A simple pendulum of mass ' m ', length ' l and charge '+q' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be



(1) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

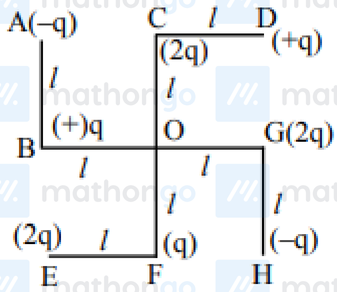
(2) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

(3) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

(4) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

Q8 (27 July 2021 Shift 2)

What will be the magnitude of electric field at point O as shown in figure? Each side of the figure is l and perpendicular to each other?



(1) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

(2) $\frac{1}{4\pi\varepsilon_0}\frac{q}{(2l^2)}(2\sqrt{2}-1)$

(3) $\frac{q}{4\pi\varepsilon_0(2l)^2}$

(4) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{2l^2}(\sqrt{2})$

Answer Key

Q1 (2)	**Q2** (4)	**Q3** (4)	**Q4** (6)
Q5 (3)	**Q6** (3)	**Q7** (3)	**Q8** (2)

Q1

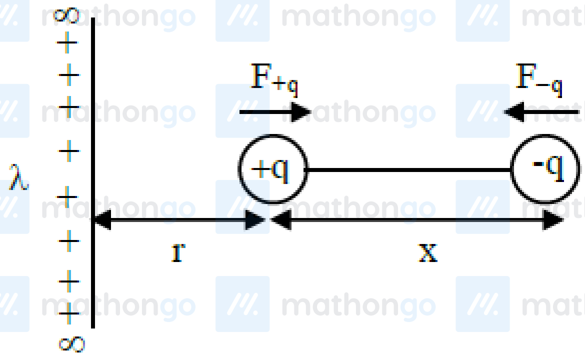
$F_q = \frac{kq(Q-q)}{L^2} = \frac{k}{L^2}\left(qQ - q^2\right)$

$\frac{dF}{dq} = 0$ when force is maximum

$\frac{dF}{dq} = \frac{k}{L^2}[Q - 2q] = 0$

$\Rightarrow Q - 2q = 0 \Rightarrow Q = 2q$

Q2



$r = 10\text{ mm}, x = 2,$

$$\left|\vec{F}_q\right| = \frac{2k\lambda}{r}\cdot q$$

$$\left|\vec{F}_{-q}\right| = \frac{2k\lambda}{r+x}\cdot q$$

$$\Rightarrow \quad \left|\vec{F}_{net}\right| = \frac{2k\lambda q}{r} - \frac{2k\lambda q}{r+x}$$

$$\left|\vec{F}_{net}\right| = \frac{2k\lambda q\cdot x}{r(r+x)}$$

$$4 = \frac{2\times 9\times 10^9\times 3\times 10^{-6}\times q\times 2\text{ mm}}{10\text{ mm}.12\text{ mm}}$$

$$\Rightarrow \quad q = 4.44\mu C$$

Q3

Electric flux density

$(\overrightarrow{\mathrm{D}})=\frac{\text{charge}}{\text{Area}}\times\hat{\mathrm{r}}=\frac{\mathrm{Q}}{4\pi\mathrm{r}^2}\hat{\mathrm{r}}=\epsilon_0\left(\frac{\mathrm{Q}}{4\pi\epsilon_0\mathrm{r}^2}\hat{\mathrm{r}}\right)$

$\Rightarrow\vec{E}=\frac{\vec{D}}{\epsilon_0}=\frac{e^{-x}\sin y\hat{i}-e^{-x}\cos\hat{y}+2z\hat{k}}{\epsilon_0}$

Also by Gauss's law

$\frac{\rho}{\epsilon_0}=\left(\frac{\partial}{\partial\mathrm{x}}\hat{\mathrm{i}}+\frac{\partial}{\partial\mathrm{y}}\hat{\mathrm{j}}+\frac{\partial}{\partial\mathrm{z}}\hat{\mathrm{k}}\right)\cdot\overrightarrow{\mathrm{E}}=\left(\frac{\partial}{\partial\mathrm{x}}\hat{\mathrm{i}}+\frac{\partial}{\partial\mathrm{y}}\hat{\mathrm{j}}+\frac{\partial}{\partial\mathrm{z}}\hat{\mathrm{k}}\right)\cdot\frac{\overrightarrow{\mathrm{D}}}{\epsilon_0}$

$\Rightarrow\rho=\frac{\partial}{\partial x}(e^{-x}\sin y)+\frac{\partial}{\partial y}(-e^{-x}\cos y)+\frac{\partial}{\partial z}(2z)$

$\rho=-e^{-x}\sin y+e^{-x}\sin y+2$

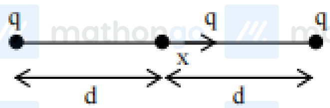
At origin $\rho=-e^{-0}\sin 0+e^{-0}\sin 0+2$

$\rho=2\mathrm{C}/\mathrm{m}^3$

Charge $=\rho\times$ volume $=2\times2\times10^{-9}=4\times10^{-9}=$

$4nC$

Q4



Net force on free charged particle $\mathrm{F}=\frac{\mathrm{kq}^2}{(\mathrm{d}+\mathrm{x})^2}-\frac{\mathrm{kq}^2}{(\mathrm{d}-\mathrm{x})^2}$

$\mathrm{F}=-\mathrm{kq}^2\left[\frac{4\,\mathrm{dx}}{\left(\mathrm{d}^2-\mathrm{x}^2\right)^2}\right]$

$a=-\frac{4kq^2d}{m}\left(\frac{x}{d^4}\right)$

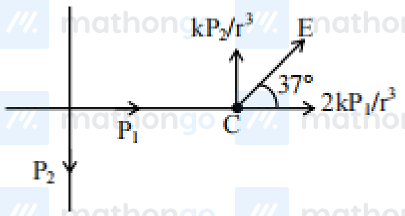
$\mathrm{a}=-\left(\frac{4\mathrm{kq}^2}{\mathrm{md}^3}\right)\mathrm{x}$

So, angular frequency $\omega=\sqrt{\frac{4\mathrm{kq}^2}{\mathrm{md}^3}}$

$\omega=\sqrt{\frac{4\times9\times10^9\times10}{1\times10^{-6}\times1^3}}$

$\omega=6\times10^8\mathrm{rad/sec}$

Q5



$\tan 37^\circ = \frac{3}{4} = \frac{\frac{kP_2}{r^3}}{\frac{2kP_1}{r^3}} = \frac{P_2}{2P_1} = \frac{3}{4}$

$\frac{P_2}{P_1} = \frac{3}{2}$

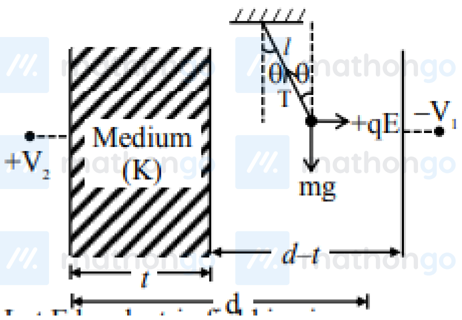
$\frac{P_1}{P_2} = \frac{2}{3}$

Q6

$V = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$

$= 3.33 \times 10^7 \ \mathrm{m/sec}$

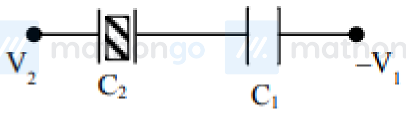
Q7



Let E be electric field in air $T \sin\theta = qE$

$T\cos\theta = mg$

$\tan\theta = \frac{qE}{mg}$



$Q = \left[\frac{C_1C_2}{C_1+C_2}\right][V_1 + V_2]$

$E=\frac{Q}{A\epsilon_o}=\left[\frac{C_1C_2}{C_1+C_2}\right]\frac{[V_1+V_2]}{A\epsilon_o}$

$C_1=\frac{\epsilon_o A}{d-t}\Rightarrow E=\frac{C_2[V_1+V_2]}{(C_1+C_2)(d-t)}$

Now $\theta=\tan^{-1}\left[\frac{q\cdot E}{mg}\right]$

$\theta=\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

Q8

$E_1=\frac{kq}{\ell^2}=E_2$

$E_3=\frac{kq}{(\sqrt{2}\ell)^2}=\frac{kq}{2\ell^2}$

$E=\frac{\sqrt{2}kq}{\ell^2}-\frac{kq}{2\ell^2}=\frac{kq}{2\ell^2}(2\sqrt{2}-1)$